

_FDSN metadata QC

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Number of stations

_FDSN

- virtual network by NSF SAGE/EarthScope (IRIS)
- FDSN backbone BB stations

MDA: 206 stations

IRISWS fedcatalog Web Service: 206 stations

http://service.iris.edu/irisws/fedcatalog/1/query?net=_FDSN&format=text&level=station

ObsPy, RoutingClient("iris-federator"): **199 stations**

Number of stations

ObsPy, RoutingClient("iris-federator"): 199 stations

```
[5]: # _FDSN
rclient = RoutingClient("iris-federator")
inv_fdsn = rclient.get_stations(network="_FDSN", level="station")
print(inv_fdsn)
```

Inventory created at 2025-08-30T19:29:45.168579Z

Created by: ObsPy 1.4.2

<https://www.obspy.org>

Sending institution: IRIS-DMC,NCEDC,ObsPy FDSN Routing 1.4.2,RESIF-SI,SeisComP,eXistDB (ANU-RSES,GFZ,INGV-ONT,IRIS-DMC,NCEDC,ODC 1),RESIF-DC)

Contains:

Networks (19):

AU, BK, CD, CN, CZ, G (2x), GE (2x), GT, H2, IC, II, IU (2x), MN,
NL (2x), PS

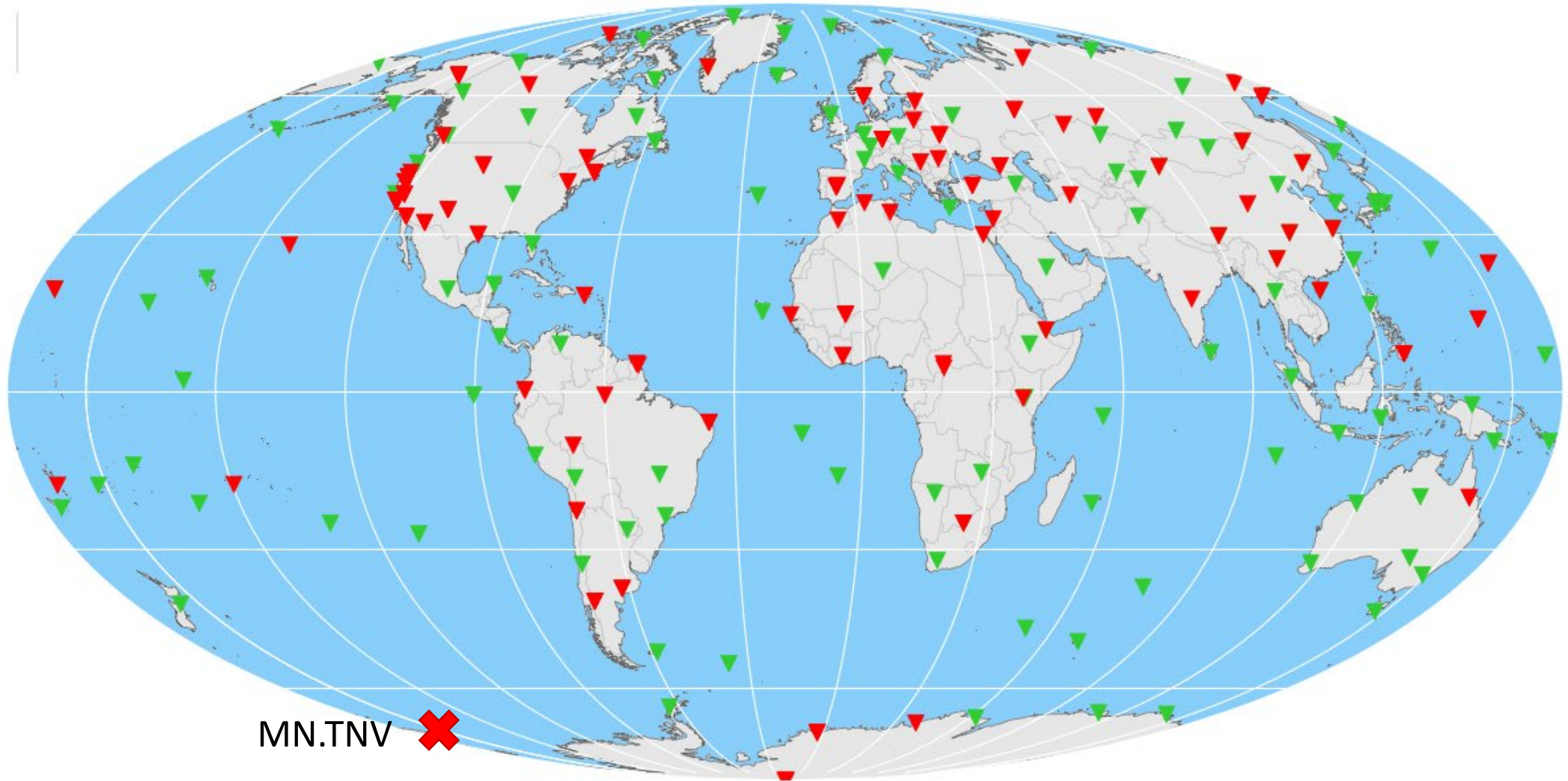
Stations (319):

AU.MAW (Mawson, Antarctica)
AU.STKA (Stephens Creek, NSW)
BK.CMB (Columbia College, Columbia, CA, USA) (2x)
BK.HUMO (Hull Mountain, OR, USA)
BK.MCCM (Marconi Conference Center, Marshall, CA, USA)
BK.SAO (San Andreas Geophysical Obs., Hollister, CA, USA) (45x)
BK.WDC (Whiskeytown Dam, Whiskeytown, CA, USA)
BK.YBH (Yreka Blue Horn Mine, Yreka, CA, USA) (20x)

net = CI is missing!

Already closed stations

- ▼ already closed
- ▼ existing up to now



MN.TNV X

IDEA: find new stations with as close locations to the old ones as possible

Metadata test

- collection of metadata tests
- first used on AdriaArray metadata, see <https://github.com/PetrColinSky/AdriaArray>
- _FDSN metadata downloaded from EIDA & IRIS on Aug 30, 2025

TESTS

- FDSN StationXML validation
- IRIS StationXML Validator - <https://github.com/iris-edu/StationXML-Validator/wiki>
 - validation levels: network, station, channel, response
- corner periods computed from the metadata responses
- checks for P&Z consistency
- cross-checks of corner periods from metadata and channel naming (band code)

List of tested issues

	id	severity		
FDSN StationXML validation	2	-1	META_VALID	# metadata FDSNStationXML validation
	2.01	3	META_ATTRIB_MISS	# metadata - missing attribute
	2.02	1	META_CLOCK_VALUE	# metadata - wrong value format in ClockDrift
IRIS StationXML validator	3.111	3	META_IRIS_111	# metadata - Station epochs overlap
	3.112.1	1	META_IRIS_112.1	# metadata - Network and/or Station endDates set to future
	3.112.2	3	META_IRIS_112.2	# metadata - Station startDate before Network startDate
	3.212.2	3	META_IRIS_212.2	# metadata - Channel startDate before Station startDate
	3.222	4	META_IRIS_222	# metadata - Station and Channel Latitudes/Longitudes differ too much
	3.223	4	META_IRIS_223	# metadata - Station and Channel Elevations differ too much
	3.304	1	META_IRIS_304	# metadata - no Sensor Description in Channel
	3.310	3	META_IRIS_310	# metadata - Channel.startDate should exist and be before Channel.endDate
	3.320	4	META_IRIS_320	# metadata - missing Azimuth and/or Dip
	3.321	4	META_IRIS_321	# metadata - invalid InputUnits (first Stage) and/or OutputUnits (last Stage)
	3.332	1	META_IRIS_332	# metadata - mis-orientation >= 5, component code should be one of [123]
	3.402.1	0	META_IRIS_402.1	# metadata - case inconsistency in Input/OutputUnits
	3.402.2	3	META_IRIS_402.2	# metadata - invalid Input/OutputUnits
	3.402.3	4	META_IRIS_402.3	# metadata - invalid InputUnits in the first Stage
	3.403	4	META_IRIS_403	# metadata - Input/OutputUnits do not follow in Stages
	3.404	5	META_IRIS_404	# metadata - missing Decimation and/or StageGain in Stage(s)
	3.410	4	META_IRIS_410	# metadata - missing Value in InstrumentSensitivity
	3.411	3	META_IRIS_411	# metadata - InstrumentSensitivity.Frequency must be < SampleRate/2
	3.414	4	META_IRIS_414	# metadata - StageGain.Frequency is zero in undue Stage
	3.416	4	META_IRIS_416	# metadata - missing InstrumentSensitivity in Response
	3.420	3	META_IRIS_420	# metadata - missing at least one Decimation in Stages
	3.421	4	META_IRIS_421	# metadata - output samplerate from Stages not equal to Channel.SampleRate
	3.422	4	META_IRIS_422	# metadata - samplerates do not follow in Stages
Response checks, corner periods, band codes	4	5	META_RESP_FAIL	# metadata response failure
	4.01	5	META_NO_POLES	# metadata - no Poles in PolesZerosResponseStage
	4.02	5	META_LAST_STAGE_FAIL	# metadata - the last response stage is faulty
	4.03	3	META_REAL_POLE_POSITIVE	# metadata - real(Pole) >= 0 in the analog PolesZerosResponseStage
	4.04	5	META_POLES_NOT_CONJUG	# metadata - not conjugated Poles in the PolesZerosResponseStage
	4.05	3	META_BAND_SAMPL_RATE	# metadata - incorrect channel band due to sampling rate
	4.06	4	META_BAND_BB_SP	# metadata - channel band confusion in broadband and short-period
	4.07	2	META_CORNER_TABLE	# metadata - corner confusion between real and tabled values

Issue severity levels

NOTIFICATIONS

0

1

not important at all

WARNINGS

2

3

station operators could
check their metadata

ERRORS

4

5

station operators **should**
check their metadata

- case inconsistency in Input/OutputUnits or CalibrationUnits

Lvl 0 is not shown
in our tests!

- wrong value format in ClockDrift
- Network and/or Station endDates set to future
- no Sensor Description in Channel
- **mis-orientation** ≥ 5 , **component code** should be one of [123] ?

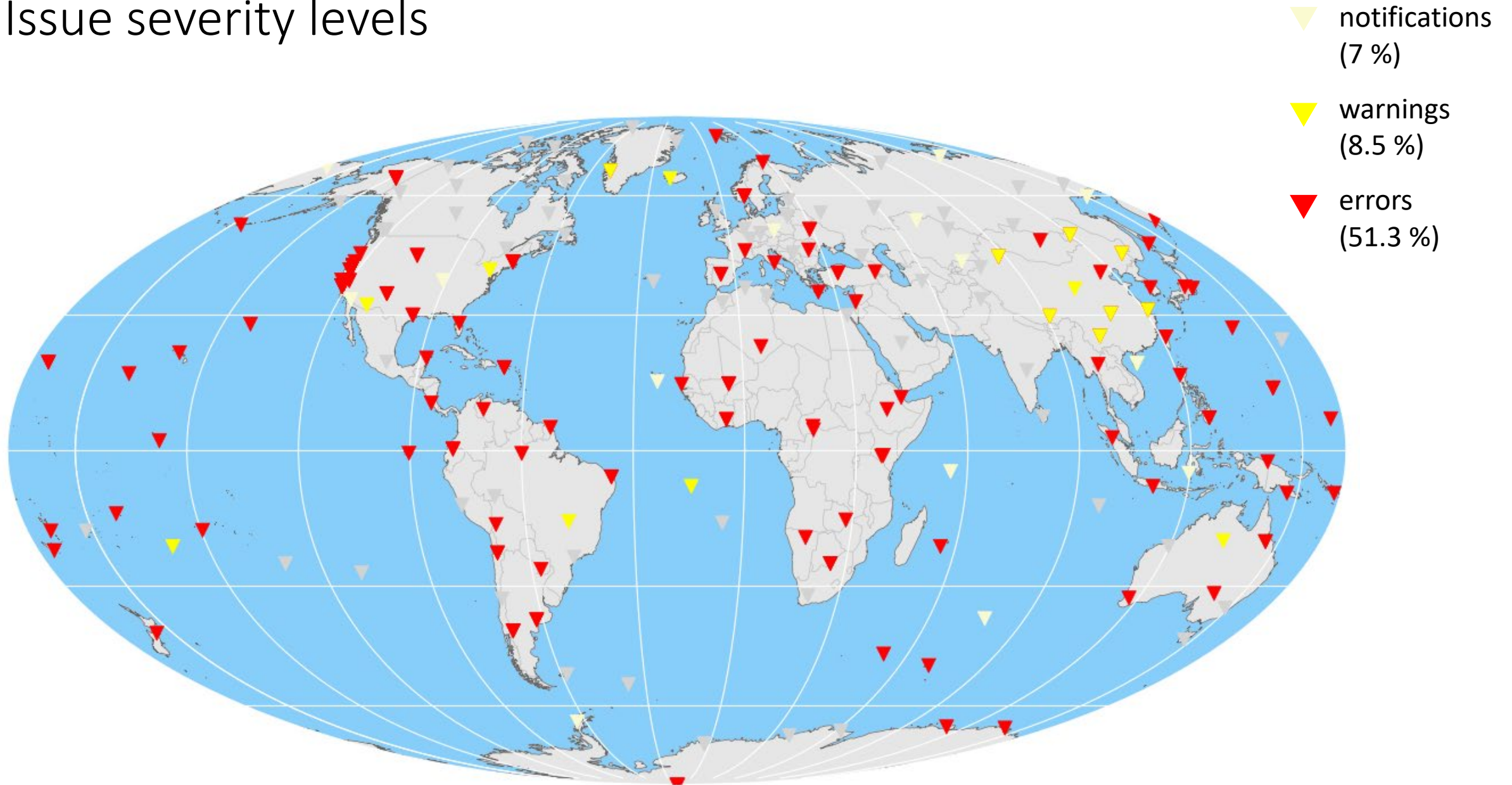
- confusion between real and tabled values of corner periods

- Station epochs overlap
- Station startDate before Network startDate
- invalid Input/OutputUnits
- missing at least one Decimation in Stages
- incorrect channel band due to sampling rate

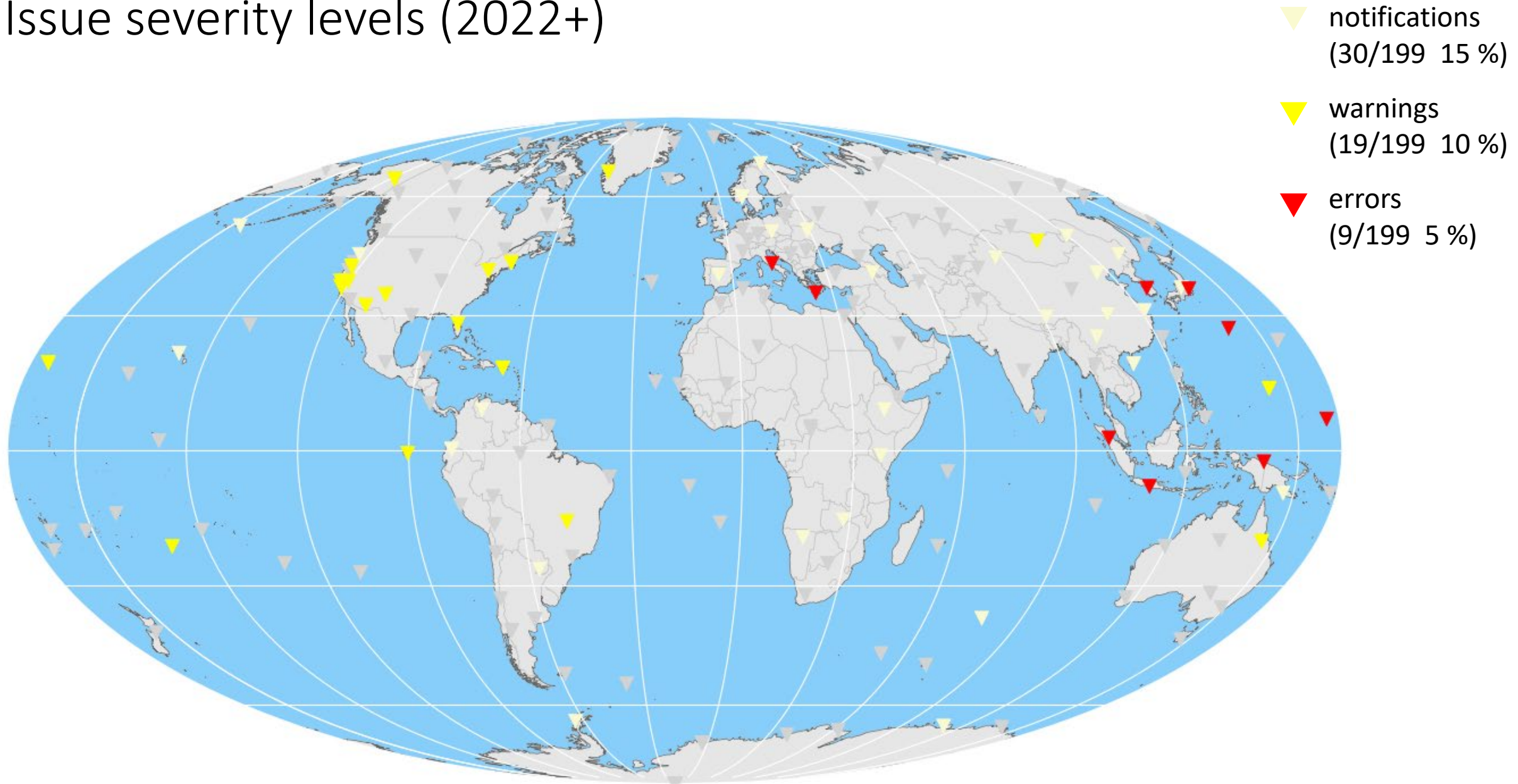
- Station and Channel Latitudes/Longitudes differ too much
- missing Azimuth and/or Dip
- missing Value in InstrumentSensitivity
- missing InstrumentSensitivity in Response
- channel band confusion in broadband and short-period

- missing Decimation and/or StageGain in Stage(s)
- metadata response failure
- no Poles in PolesZerosResponseStage
- the last response stage is faulty
- not conjugated Poles in the PolesZerosResponseStage

Issue severity levels



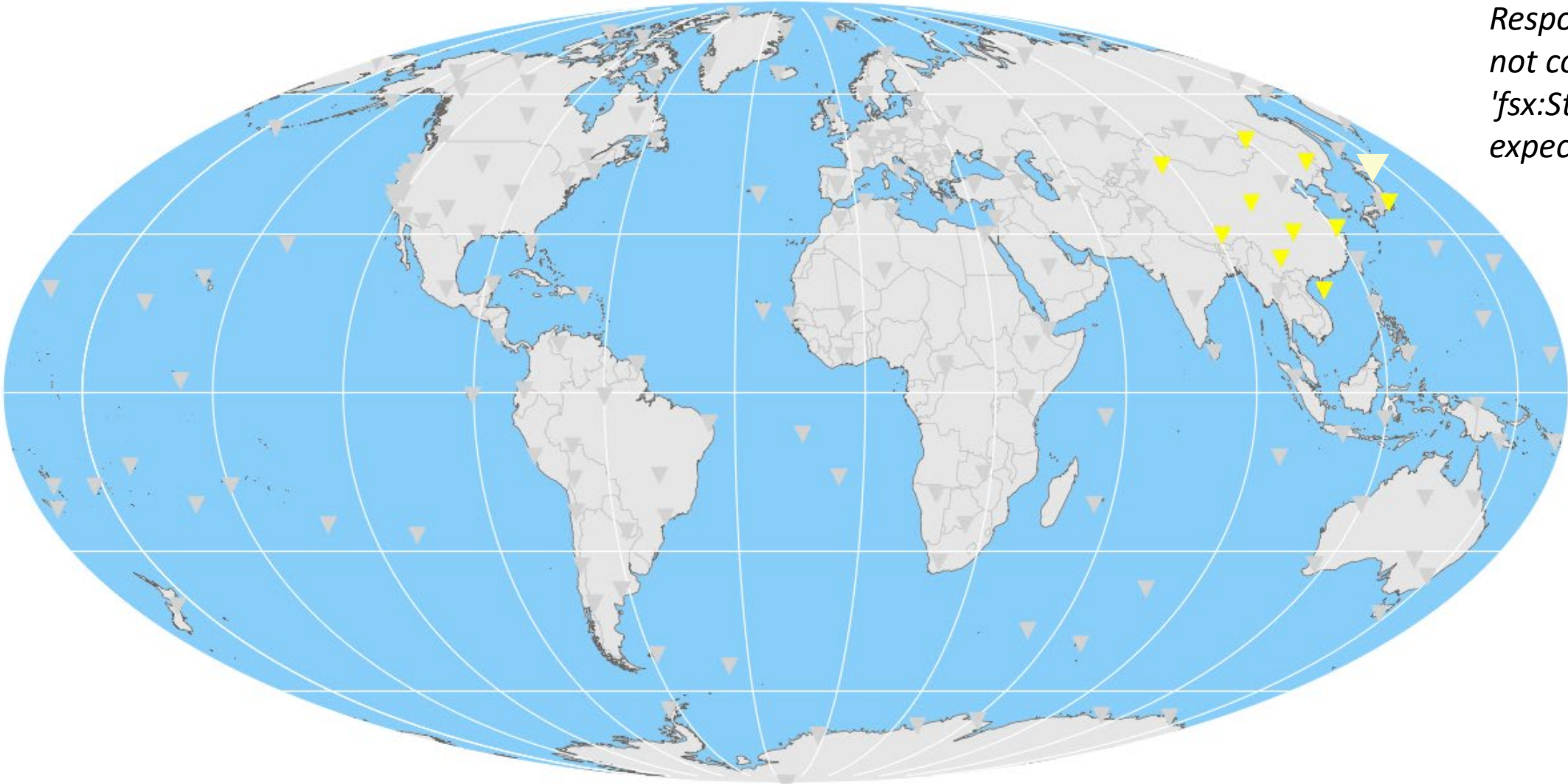
Issue severity levels (2022+)



FDSN StationXML validation

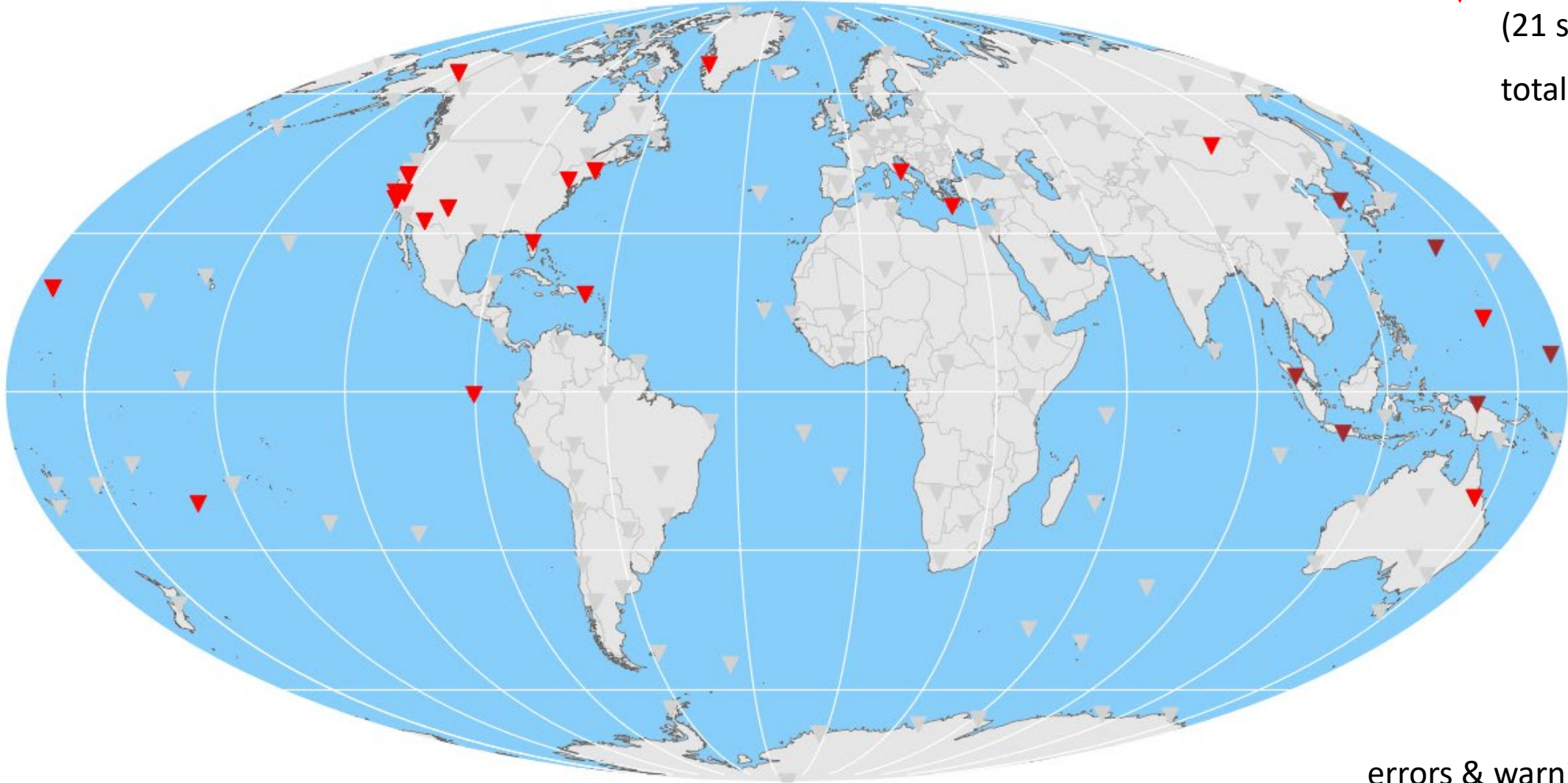
▼ missing attribute
(10 stations)

*ResponseStage is
not complete. Tag
'fsx:StageGain'
expected*



IRIS StationXML validator without notifications

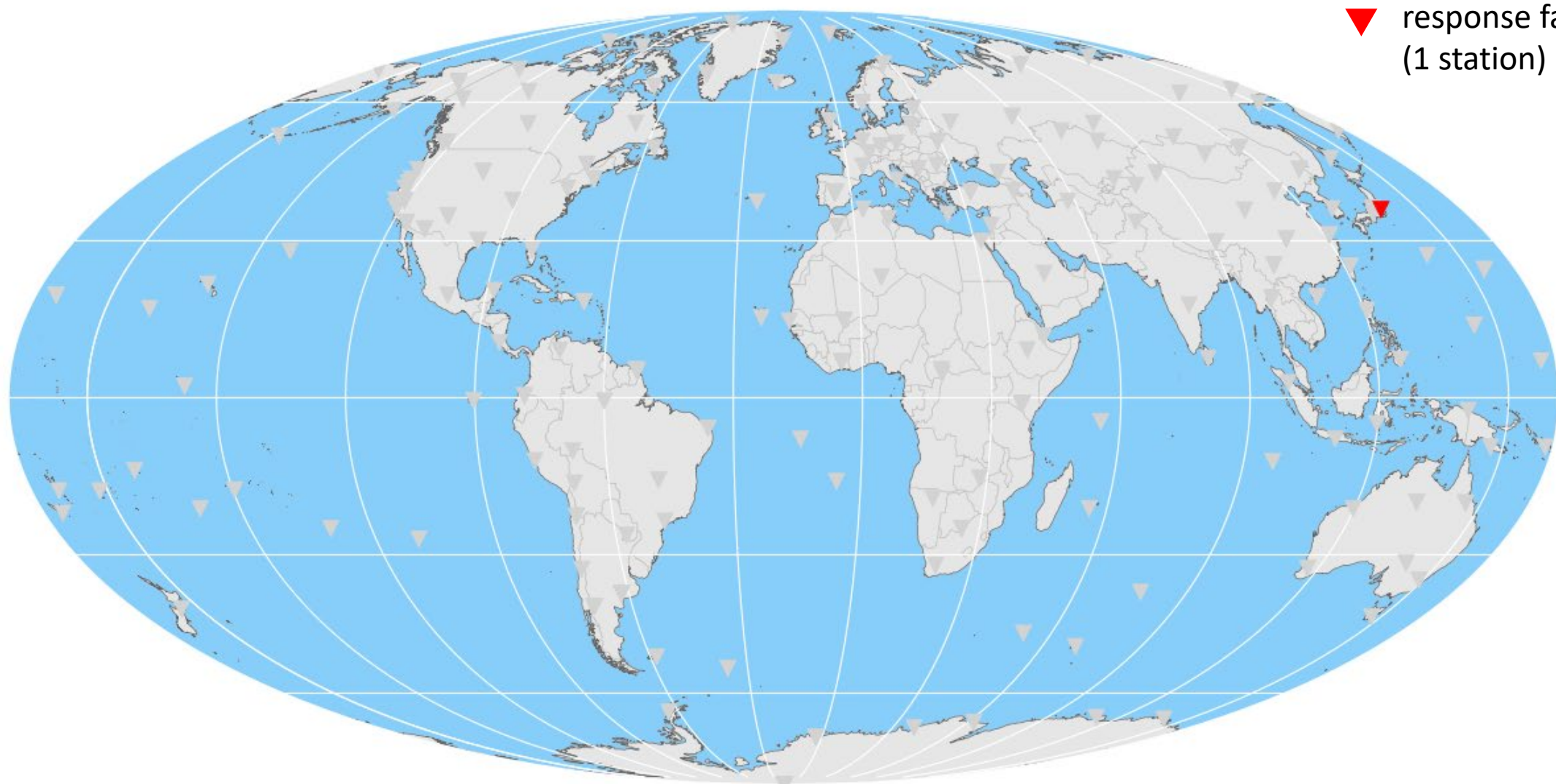
- ▼ channel level
(6 stations)
- ▼ response level
(21 stations)
- total: 14 %



errors & warnings
only 2022+

Another response checks

errors
only 2022+



PS.TSK..MHE;1989-09-15;metadata response failure;META:IRIS-DMC:250830;found:250830;must be real number, not NoneType

Corner period, band codes

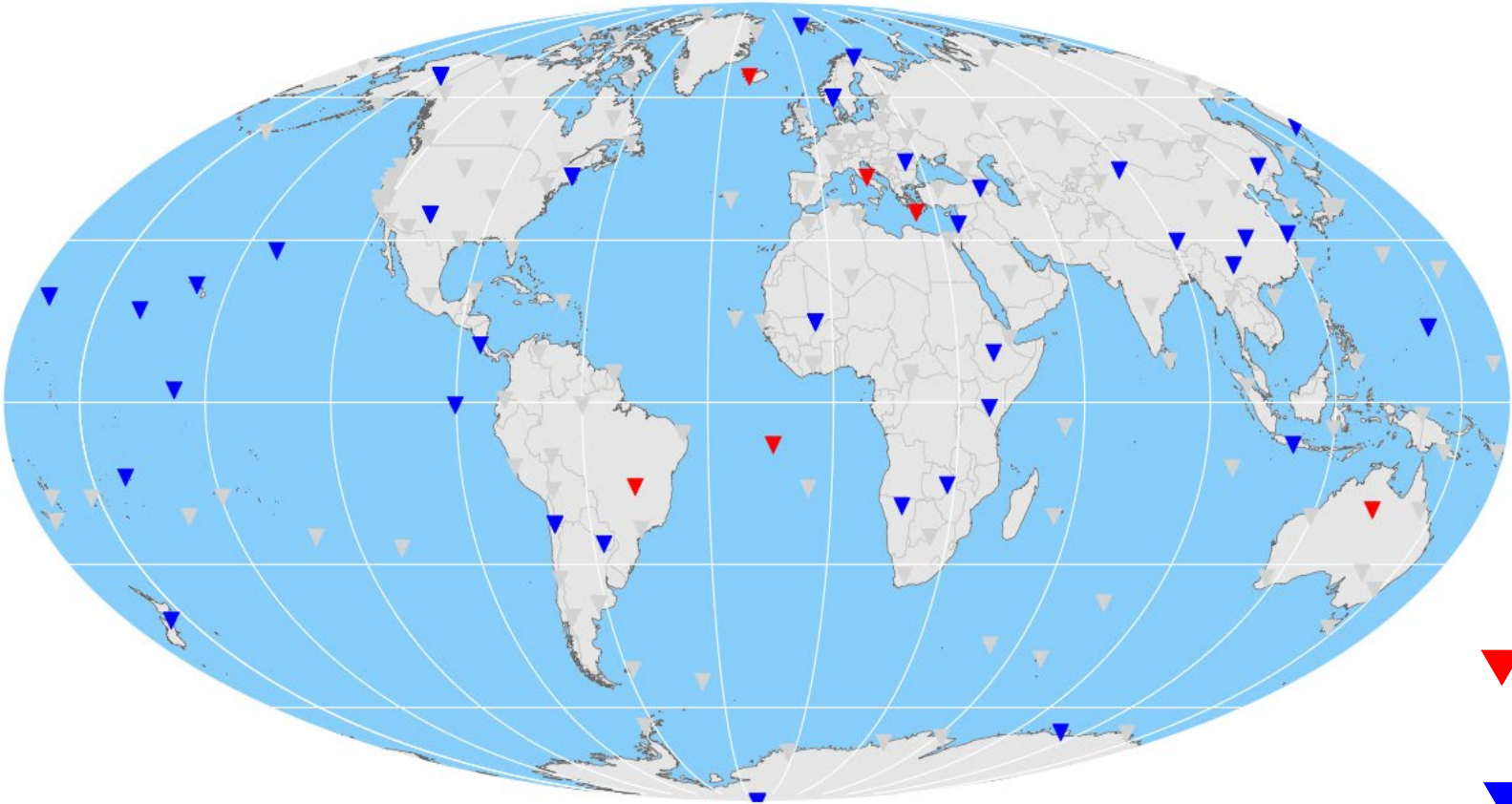
sampling
frequency
20 Hz

+

corner
period
60 s

=

band
code
BHZ



Band Code			
The first letter specifies the general sampling rate and the response band of the instrument. (The "A" code is reserved for administrative functions such as miscellaneous state of health.)			
Band code	Band type	Sample rate (Hz)	Corner period (sec)
F	...	≥ 1000 to < 5000	≥ 10 sec
G	...	≥ 1000 to < 5000	< 10 sec
D	...	≥ 250 to < 1000	< 10 sec
C	...	≥ 250 to < 1000	≥ 10 sec
E	Extremely Short Period	≥ 80 to < 250	< 10 sec
S	Short Period	≥ 10 to < 80	< 10 sec
H	High Broad Band	≥ 80 to < 250	≥ 10 sec
B	Broad Band	≥ 10 to < 80	≥ 10 sec
M	Mid Period	> 1 to < 10	
L	Long Period	≈ 1	
V	Very Long Period	≈ 0.1	
U	Ultra Long Period	≈ 0.01	
R	Extremely Long Period	≥ 0.0001 to < 0.001	
P	On the order of 0.1 to 1 day ¹	≥ 0.00001 to < 0.0001	
T	On the order of 1 to 10 days ¹	≥ 0.000001 to < 0.00001	
Q	Greater than 10 days ¹	< 0.000001	
A	Administrative Instrument Channel	variable	NA
O	Opaque Instrument Channel	variable	NA

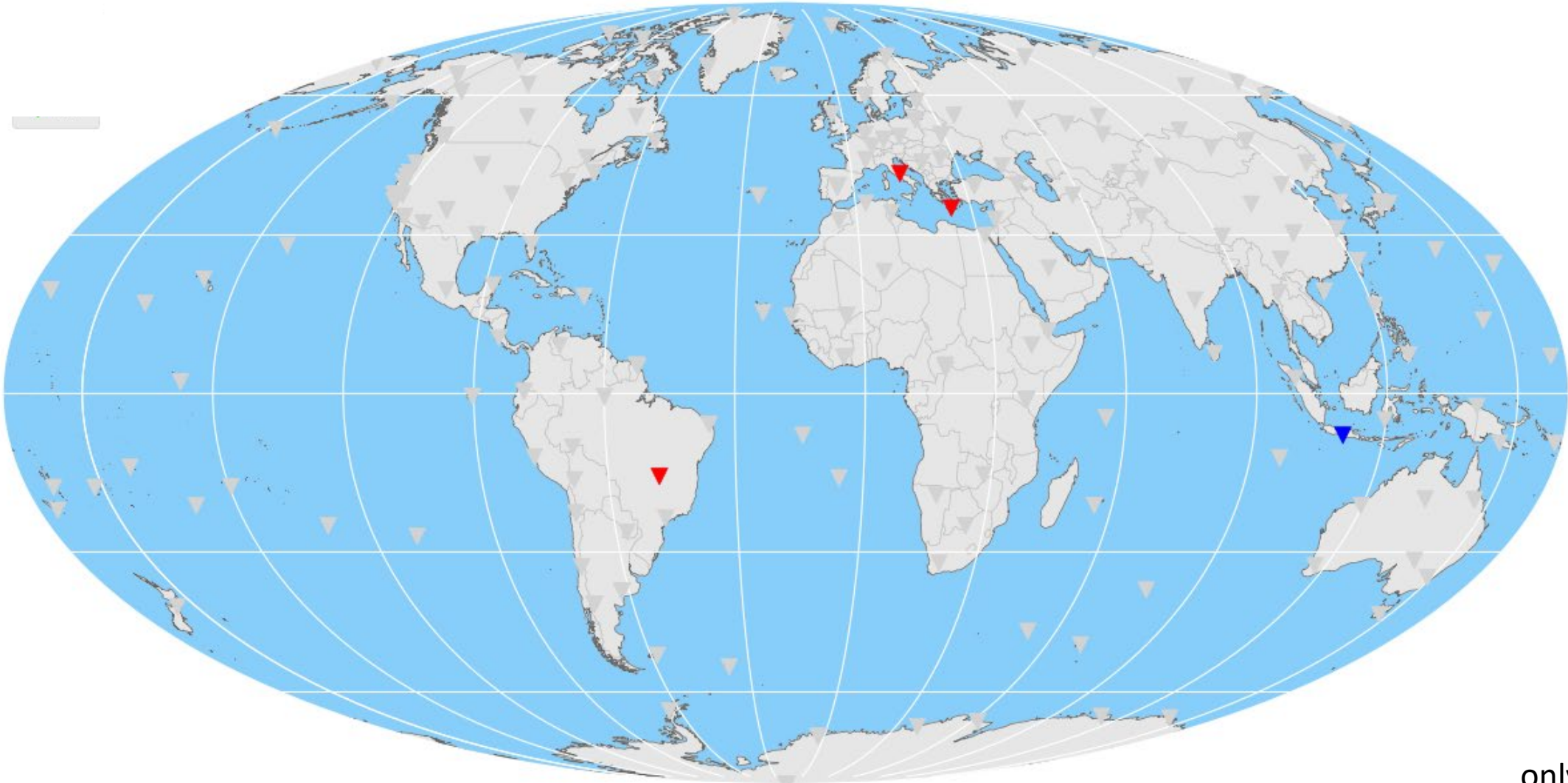
1. These are approximate values. The sample rate should be used for the correct Band Code.

[SEED manual](#)

- ▼ wrong band code for a given sample rate (6 stations)
- ▼ wrong band code for broad-band/short-period (37 stations)

Corner period, band codes

- ▼ wrong band code for a given sample rate (3 stations)
- ▼ wrong band code for broad-band/short-period (1 station)



only 2022+

Wrong band code for a given sample rate

GT.BDFB..BH[Z12]	2009-01-14 to 2022-09-19	band code B -> H (sr: 100.0 Hz, corner period: 119.80 s)
MN.AQU..VH[ZNE]	2019-03-01 to now	band code V -> L (sr: 1.0 Hz, corner period: 128.38 s)
MN.IDI..BH[ZNE]	2017-12-19 to now	band code B -> H (sr: 80.0 Hz, corner period: 125.46 s)
MN.IDI..LH[ZNE]	2017-12-19 to now	band code L -> H (sr: 80.0 Hz, corner period: 125.46 s)

Wrong band code for broad-band/short-period

GE.UGM..SH[ZNE]	2006-06-29 to now	band code S -> B (sr: 50.0 Hz, corner period: 122.60 s)
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Concluding remarks

- We can identify issues in station metadata
- Next step #1: results should be published (csv tables, figures)
- Next step #2: contact operators and/or
<https://github.com/EIDA/userfeedback/issues>